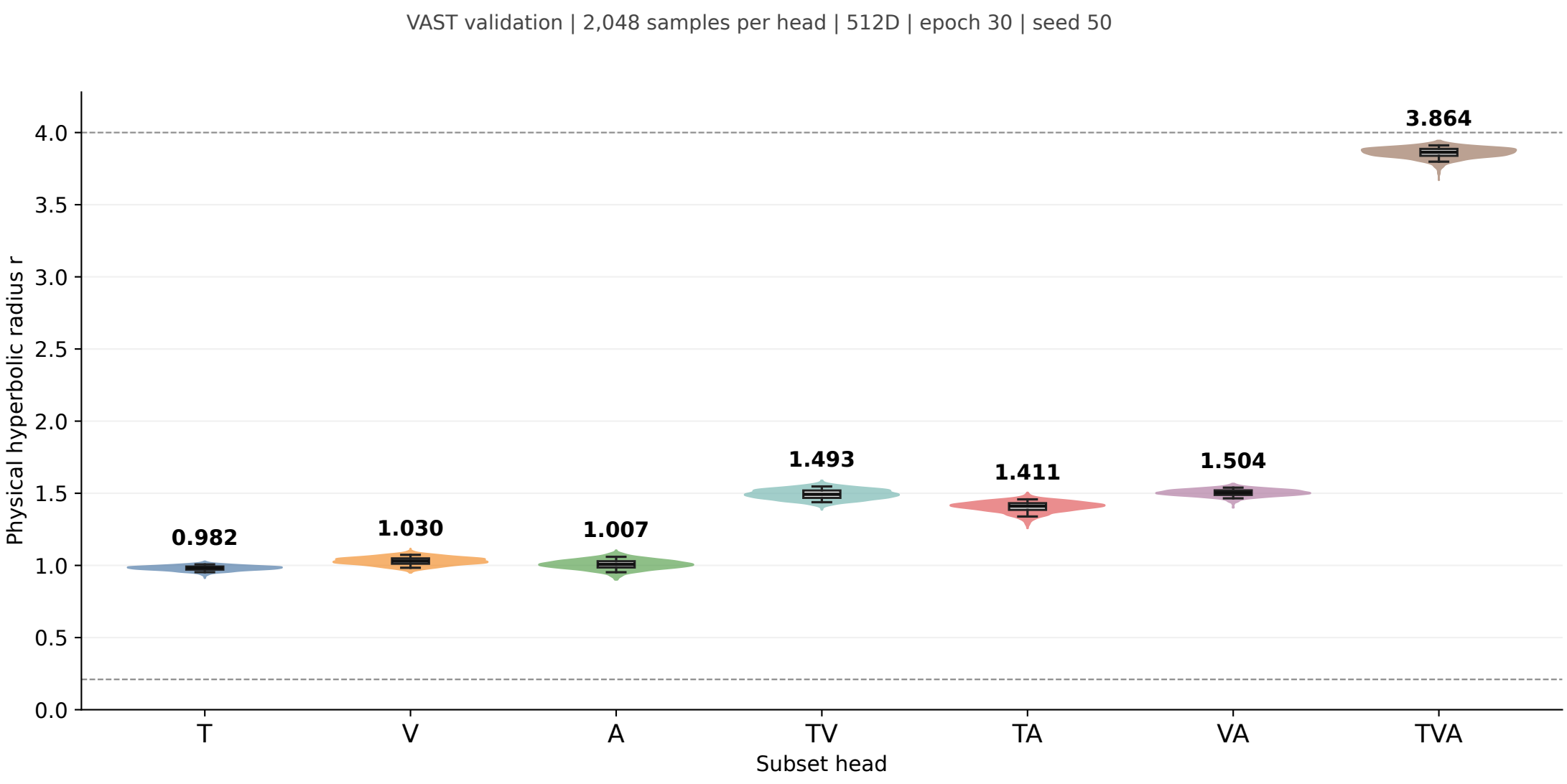
Fixed c = 1, tau = 0.035 | lambda = 0.2
c and tau fixed throughout training
VAST validation | 2,048 samples per head | 512D | epoch 30 | seed 50
Physical hyperbolic radius r
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0.982
1.030
1.007
1.493
1.411
1.504
3.864
T
V
A
TV
TA
VA
TVA
Subset head
Violin: sample density | Box: 25–75% | Center line: median | Whiskers: 5–95%
Numbers above boxes: median | Dashed lines: model radius bounds (0.21 and 4)